

Spring 2021 Real Analysis II — Homework 4

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1. Let X and Y be Banach spaces. Show that $T \in L(X, Y)$ if and only if $T : X \rightarrow Y$ is linear and $T(U) \subset Y$ is bounded whenever $U \subset X$ is bounded.
2. Let X and Y be Banach spaces and $T \in L(X, Y)$. Show that

- (a) $\|T\| = \sup_{\|x\| \leq 1} \|Tx\|$.
- (b) $\|T\| = \sup_{\|x\| < 1} \|Tx\|$.
- (c) $\|T\| = \inf\{M \in \mathbb{R} : \frac{\|Tx\|}{\|x\|} \leq M, \forall x \in X \setminus \{0\}\}$.

3. Let X be a Banach space and $f \in X^*$. For any $x_0 \in X$ and $\delta > 0$, show that

- (a) $\sup_{x \in B(x_0; \delta)} f(x) = f(x_0) + \delta \|f\|$.
- (b) $\inf_{x \in B(x_0; \delta)} f(x) = f(x_0) - \delta \|f\|$.

4. Let X be a Banach space and $f \in X^*$ be nonzero. Show that there exists no open ball $B(x_0; \delta)$ such that $f(x_0)$ is an extreme value of f on $B(x_0; \delta)$.
5. Let $X = L^1([a, b])$ and $Y = C([a, b])$. Define $T : X \rightarrow Y$ by

$$(Tf)(x) = \int_a^x f(t) dt, \quad \forall f \in X.$$

- (a) Find the norm of $T \in L(X, Y)$.
 - (b) Find the norm of $T \in L(X, X) =: L(X)$.
6. Let X be a Banach space and $f \in X^*$. Show that for any $\epsilon > 0$ there exists $x_0 \in X$ such that $f(x_0) = \|f\|$ and $\|x_0\| \leq 1 + \epsilon$.
 7. Let X and Y be Banach spaces and $T : X \rightarrow Y$ be linear. Denote $Z(T) := \{x \in X : Tx = 0\}$.
 - (a) Suppose $T \in L(X, Y)$. Show that $Z(T)$ is a closed linear subspace of X .
 - (b) Suppose $Y = \mathbb{R}$. Show that $T \in X^*$ if and only if $Z(T)$ is a closed linear subspace of X .
 8. Let X be a Hilbert space and $f \in X^*$. Suppose $C \subset X$ is a closed convex set and $g : C \rightarrow \mathbb{R}$ is defined by $g(x) := \frac{1}{2}\|x\|^2 - f(x)$ for any $x \in C$. Show that
 - (a) There exists a unique $y \in X$ such that $g(x) = \frac{1}{2}\|x - y\|^2 - \frac{1}{2}\|y\|^2$ for any $x \in C$.
 - (b) There exists a unique $z \in C$ such that $g(z) = \inf_{x \in C} g(x)$.

9. Let S be a set and $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space, where the elements of X are real-valued functions defined on S . For any $x \in S$, define $J_x \in X^*$ by $J_x(f) := f(x)$. Show that there exists $K : S \times S \rightarrow \mathbb{R}$ such that

- (a) For any $y \in S$, $K(\cdot, y) \in X$ is a real-valued function defined on S .
- (b) For any $f \in X$ and $y \in S$, there is $f(y) = \langle f, K(\cdot, y) \rangle$.

Such function K is called a *reproducing kernel of the Hilbert space* X , and X is called a *reproducing kernel Hilbert space (RKHS)*.

10. Let S and X be defined as above, and $\{e_k : k \in \mathbb{N}\}$ be a basis of X . Suppose K is a reproducing kernel. Show that $K(x, y) = \sum_{k=1}^{\infty} e_k(x)e_k(y)$ for all $x, y \in S$.